

The Execution Reality Gap

Candidate Assignment, Execution Module

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1 Part A. Friction models and the slippage waterfall

Friction levels stack on one trade stream, each subtracting cost from the Level 0 baseline (all randomness seeded). ATR(14) uses a simple moving average rather than Wilder’s RMA, the only computational choice left open by the assignment. Levels 0 and 1 follow the specification directly. Levels 2 and 3 required design choices, justified below.

Level 2, Slippage + latency.

Each order draws a master execution latency from a lognormal (median 0.5 s, $\sigma = 0.5$), chosen because execution latency is strictly positive and right skewed (Cartea and Sánchez Betancourt, 2022). The median places fills in the first executable minute after the signal. The fill price is read from the real M1 path. Stops are gap-aware. They trigger when the minute path touches the level and fill at the stop or worse, never better.

Level 3, Partial fills.

Each minute fills a fraction $\min(1, \text{tickvol}_t/\widetilde{\text{tickvol}})$ of the order, so under normal liquidity L3 = L2 and tranching only bites in thin minutes. The charge is one-sided (delayed tranches always cost, never pay) because crediting favourable late fills would require passive limit orders whose non fill risk is not modelled. A symmetric variant is stress tested in the findings.

1.0.1 Observed vs. modelled (and its limits)

With minute bars the within-hour price path is largely *observed* rather than modelled. The minute in which a stop is triggered and the delayed execution bar are read directly from the M1 data. Two pieces remain *modelled*. (i) **sub-minute execution price**. Within a minute we only observe OHLC, so the execution price is linearly interpolated between the open and close at the latency’s fractional offset f . At $f \approx 0.008$ (the 0.5s median) this is negligible and becomes material only at the minute-scale delays of Part B. (ii) **partial-fill sizing**. The fill fraction per bar is $\min(1, \text{tickvol}_t/\widetilde{\text{tickvol}})$, a deliberate simplification consistent with the assignment’s simple-fill-model requirement. The global median $\widetilde{\text{tickvol}}$ spans the full sample (2020–2026) and therefore incorporates in-sample data. A rolling pre-period median is not tested and constitutes a minor look-ahead. On tick data the interpolation error would vanish, whereas partial-fill sizing would still require an execution model because queue position and realised fills remain unobserved. The ETH exit slippage would therefore likely come out larger rather than smaller.

1.1 The slippage waterfall

1.1.1 The waterfall

Figure 1 shows the cumulative decay from Level 0 $\rightarrow$ 3 per asset, in basis points of the entry notional per round-trip trade (the only cross-asset comparable unit, since price points differ in scale). Table 1 gives the numbers alongside gross to net PnL.

Asset	n	L0 (gross)	L3 (net)	total cost (bps/RT)
SPXUSD	1122	−539.4	−1621.7	2.134
USDJPY	1038	36.0	14.7	1.518
XAUUSD	1044	286.8	−245.6	1.943
ETHUSD	1029	4573.5	494.0	19.657

Table 1: Level 0 gross vs. Level 3 net PnL, asset-specific price points (comparable only *within* an asset, the price scales differ). SPXUSD is already loss-making gross, so its Level 3 loss is not an execution artefact. XAUUSD flips sign under friction alone. The bps cost decomposition behind the total column is Figure 1.

The assignment also asks for every fill’s slippage logged against its original signal timestamp, separately from the round-trip mid-drift attribution of Figure 1. Table 2 gives that per fill view.

Asset	entry slip (bps)	exit slip (bps)
SPXUSD	0.257	0.191
USDJPY	0.004	0.061
XAUUSD	−0.091	0.252
ETHUSD	0.995	9.290

Table 2: Per fill slippage logged against the original signal timestamp (mean, bps of entry notional). Entry is the delay onto the first executable price, exit carries the stop gap risk. These per fill figures do not sum to the round-trip slippage of Figure 1.

Part A - Execution reality gap: Level 0 -> 3 cost waterfall per asset

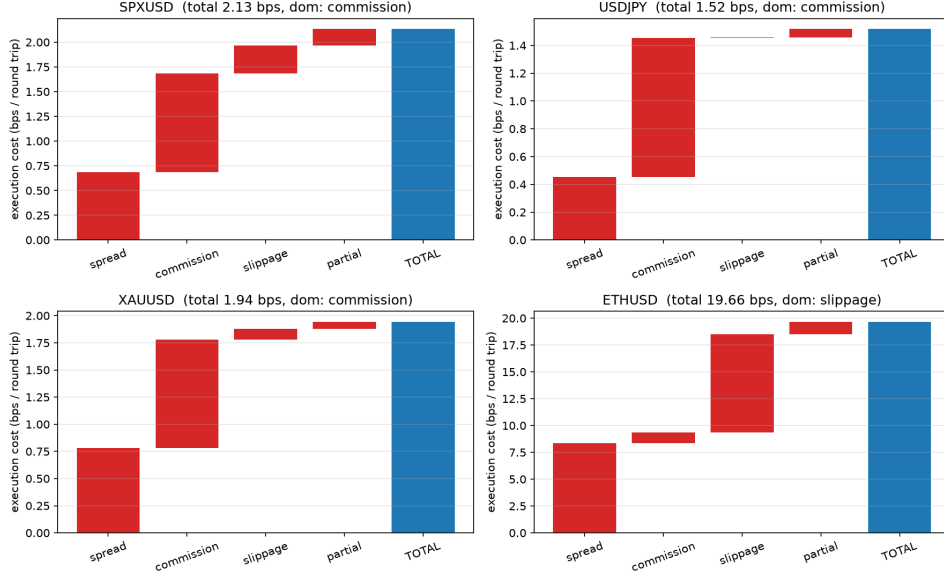


Figure 1: Execution cost waterfall per asset (bps per round trip). The dominant friction is commission on the liquid majors and slippage on crypto.

1.1.2 Findings

- **The dominant friction flips by asset class** (Table 1, Figure 1). Commission leads on the majors (1.5–2.1 bps total). Slippage leads on ETH (19.7 bps, ten times larger), the spread dispersion the dataset was built to expose.
- **Execution alone can flip a winner into a loser.** XAUUSD’s gross edge is fully erased by friction (+287 → −246 points). ETH keeps only $\approx 11\%$ of its gross edge. SPXUSD is the exception. It is already loss-making gross, so its Level 3 loss is not an execution story.
- **Latency slippage is an exit/stop phenomenon, not an entry one** (Table 2). Entry slippage is uniformly small. The exit side carries the stop gap risk and dominates on every asset except SPXUSD, most visibly on ETH, where a jumped stop fills at the gap, not the level.
- **Partial fills are a tail, not an average.** Mean cost is small (0.06–0.17 bps majors) but concentrated, non-zero in $\approx 10\%$ of trades, with the top five carrying 35–65% of the tranching cost per asset. The driver is thin liquidity (USDJPY’s worst tranches fall in the 20–26 Dec holiday week), not crash volatility. The model prices illiquidity, not impact.
- **The one-sided charge is not driving the conclusion.** A symmetric tranche variant moves the total by at most ≈ 0.5 bps and does not flip the dominant friction on any asset (commission majors, slippage ETH), the conservative penalty is a bounded, visible assumption.

2 Part B. The replication problem

2.1 Signal-side versus fill-side findings

A signal side follower fills at $t_s + d$, a fill side follower at $t_s + L_m + d$, separated by a constant $L_m \approx 0.55$ s (the master’s own latency). Table 3 measures the consequence for SPXUSD.

median (s)	lag gap (s)	signal faster	bar cross sig/fill (%)	decay sig/fill (bps)
0.3	0.552	70.6%	0.0/0.0	−0.00/0.00
30	0.552	0.0%	17.3/17.6	0.71/0.73
120	0.552	0.0%	80.9/81.3	2.74/2.75
600	0.553	0.0%	98.2/98.2	5.63/5.63

Table 3: Signal vs. fill side follower diagnostics for SPXUSD (means over 25 followers). The lag gap equals L_m by construction. Decay is incremental bps relative to the Level 2 master.

The decay gap between the two modes stays below 0.06 bps everywhere, two orders of magnitude below the main follower decay. The choice is operational (fill side avoids phantom signals) rather than PnL driven. On tick data the L_m gap would map into quote level price changes instead of only occasional bar crossings, sharpening the distinction. The remainder of Part B fixes signal side replication.

2.2 Follower performance distribution

Figure 2 and Table 4 show the cross follower distribution at the two delay levels where replication visibly matters.

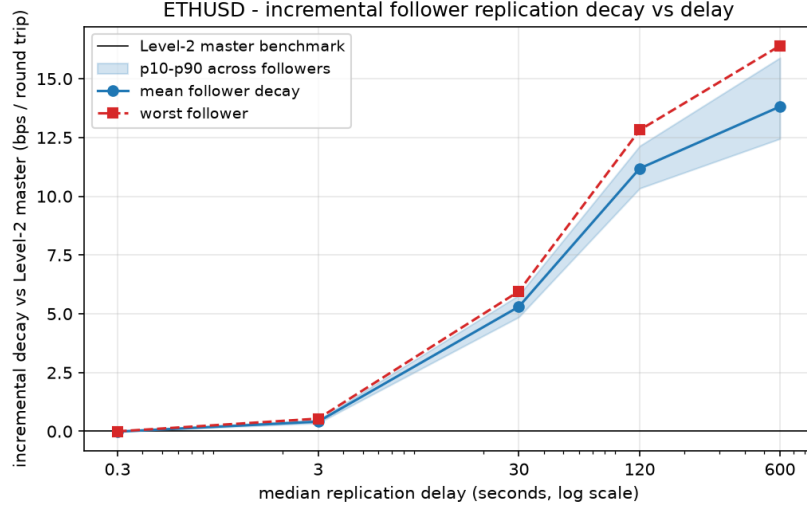


Figure 2: ETHUSD follower decay (bps) vs. the Level 2 master. The shaded band is p10–p90 across the 25 followers. Dispersion and tail deteriorate as delay reaches minute scale.

Asset	median (s)	mean	std	p10–p90	worst
SPXUSD	120	2.74	0.18	2.52–2.95	3.11
	600	5.63	0.40	5.19–6.13	6.49
USDJPY	120	1.32	0.09	1.20–1.42	1.45
	600	2.35	0.19	2.11–2.64	2.73
XAUUSD	120	1.11	0.15	0.94–1.25	1.44
	600	1.77	0.28	1.46–2.12	2.39
ETHUSD	120	11.18	0.86	10.34–12.14	12.82
	600	13.81	1.35	12.45–15.89	16.40

Table 4: Follower decay distribution (bps vs. master) at the two delays where replication matters. Signal side, 25 followers. Every follower underperforms the master in this seeded run.

The mean decay is an incomplete summary. The more important effect is the widening distribution. Followers running identical policy fan out as delay grows, and the worst follower exceeds the mean by 9–35% across the four assets, widest for XAUUSD. Replication delay is less a stable tax than a dispersion and tail risk amplifier, which is why the sizing optimisation below (Optimisation 1) is evaluated on the worst-case follower as well as the mean.

2.3 Optimisations with measured effects

All optimisation results use common random numbers, so differences are attributable to the optimisation rather than to resampling noise.

2.3.1 Optimisation 1. Volatility-normalised sizing

A delayed copy is riskier when short horizon volatility is high. Each follower entry is therefore sized as

$$q(d, \sigma_t) = \min\left(1, \frac{R^*}{d \cdot \sigma_t}\right),$$

where σ_t is the rolling standard deviation of M1 close-to-close returns (in bps) over a 60-bar window, shifted by one bar so that σ_t uses only bars strictly before the signal (look-ahead free, verified by assertion). $R^* = 30 \text{ s} \times \text{median}(\sigma_t)$ per asset, giving full size at representative volatility and a 30 s delay. The 30 s anchor comes from the bar-crossing diagnostic. At 30 s median delay the order-level bar-crossing rate is 17–18% across all four assets, confirming that “where delay starts to bite” is a property of the M1 sampling grid, not the market. R^* is therefore not fitted to PnL. Exits always close the full position.

Asset	median (s)	avg. q	before	after	improvement	worst after	full copy	sized copy
SPXUSD	120	0.216	2.74	-2.16	4.90	-1.26	-5.82	-0.92
	600	0.046	5.63	-2.78	8.41	-2.57	-8.70	-0.30
USDJPY	120	0.238	1.32	1.00	0.32	1.67	0.16	0.48
	600	0.051	2.35	1.43	0.92	1.60	-0.87	0.05
XAUUSD	120	0.221	1.11	-0.50	1.61	0.36	-1.08	0.53
	600	0.046	1.77	-0.05	1.82	0.16	-1.75	0.07
ETHUSD	120	0.223	11.18	3.99	7.19	9.74	-2.36	4.84
	600	0.047	13.81	7.99	5.83	9.26	-4.99	0.84

Table 5: Vol-normalised sizing results. Before/after/improvement and worst after are mean and worst-follower decay (bps). Full copy and sized copy are the follower’s own mean per-trade PnL (bps) at $q = 1$ and under the sizing rule. Sized copy exceeds full copy everywhere. At 600 s the fleet averages $\approx 5\%$ of master notional.

Tail exposure. Comparing the worst after column above with the worst column of Table 4, the worst-case follower’s decay falls at both delays for SPXUSD, XAUUSD and ETHUSD. USDJPY is the exception at 120 s, where the worst follower is marginally higher than before sizing ($1.45 \rightarrow 1.67$ bps).

Edge given up. The last two columns of Table 5 show that the sized copy’s PnL exceeds the full-size copy’s PnL on all four assets at both delays. No net PnL is given up. The exposure the rule removes (the $1 - q$ fraction of each trade) is net losing, so cutting it improves follower PnL and decay by the same amount by construction. The genuine cost is participation, not PnL. At avg. $q = 0.047$ the follower keeps under 10% of the master’s edge. The rule is protection against bad fills, not recovery of the edge.

2.3.2 Optimisation 2. Local stop execution

Stop exits are deterministic. The level is known once the follower has entered. Instead of waiting for the master’s replicated stop message, the follower monitors its own M1 path against the copied stop level and exits locally when breached. The post-breach fill still draws an independent execution latency, so the improvement removes only the replication delay, not the physical order-to-fill friction. Setting post-breach latency to zero instead of the independent draw moves the result by ≤ 0.003 bps across all assets and delays. 56–61% of exits are stops, so this applies to the majority of round trips. Table 6 reports the mean follower decay before and after the optimisation.

Asset	median (s)	before	after	improvement
SPXUSD	120	2.74	0.77	1.96
	600	5.63	1.18	4.45
USDJPY	120	1.32	0.22	1.10
	600	2.35	0.23	2.13
XAUUSD	120	1.11	-0.35	1.46
	600	1.77	-1.18	2.95
ETHUSD	120	11.18	5.93	5.25
	600	13.81	3.36	10.45

Table 6: Mean follower decay (bps) before and after local stop execution.

Edge given up. No trades are skipped and no positions are downscaled. The cost is model risk. Divergent exits (follower stops where the master does not, or vice versa) grow with delay but remain rare, 2.6–7.8 per follower at 600 s. They carry marginally *negative* PnL impact across all four assets, so divergence does not represent a meaningful edge given up in this sample.

XAUUSD negative decay. At long delays the after-values for XAUUSD turn negative. A per-trade diagnostic confirms the mechanism. The copied stop is anchored to the *master* entry price, so a delayed follower that enters closer to the stop has a tighter effective stop distance and loses less on stop-outs. This anchoring effect operates on $\geq 99\%$ of negative-decay trades across all four assets but only crosses zero in the aggregate where the pre-optimisation baseline was already small. XAUUSD’s 1.77 bps at 600 s is the lowest of the four. It is a structural consequence of stop inheritance, not a XAUUSD-specific artefact.

3 Part C. The risk gate

3.1 Model

Only *entry* orders pass through the gate. Exits are exempt because a gate that blocks a risk-reducing exit defeats its own purpose. On timeout Policy S trades at half size priced at τ (not the actual $g_i > \tau$, since

waiting would defeat the timeout). An informed rejection variant (lowest 2% of baseline PnL per instrument) is reported alongside the uninformed default as a robustness check. A blocked entry does not free the strategy’s state for the next signal, revisited under Sensitivity. The gate is implemented independently from the execution engine, enabling new timeout policies to be added without modifying execution logic.

3.2 Which policy dominates, and where

Under common random numbers the timed-out set T is identical for both policies. The net impact difference is $\Delta = \frac{1}{2} \sum_{i \in T} \pi_i$. Policy H removes all timeout exposure. Policy S preserves half. The entire policy difference is therefore half of the timed-out orders’ PnL, whose expected sign is simply the instrument’s edge sign, so H is preferred exactly when timed-out orders carry negative net PnL. Figure 3 and Table 7 confirm this.

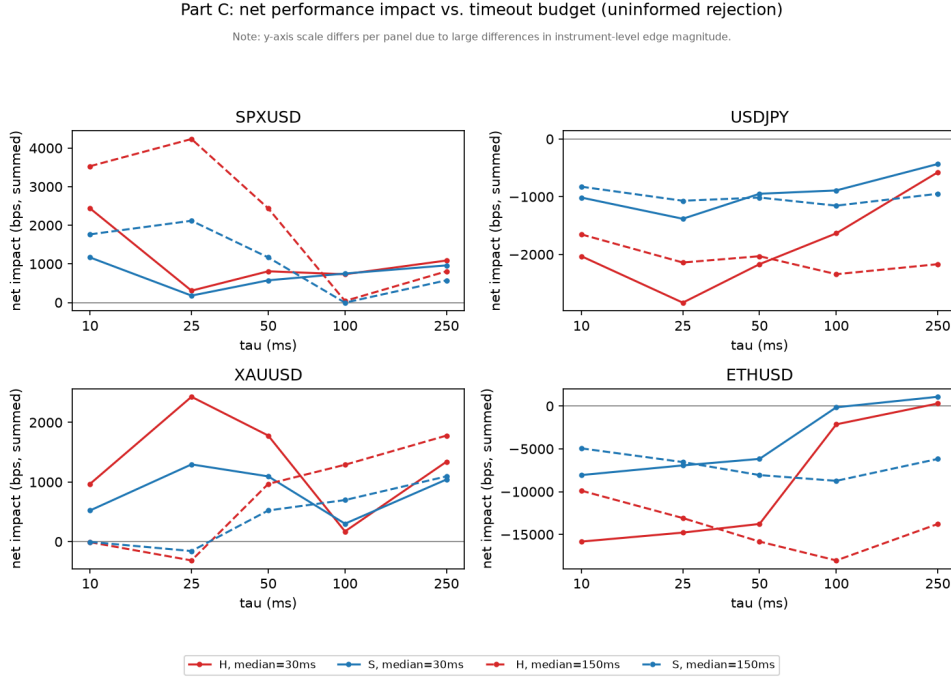


Figure 3: Net performance impact (summed bps) vs. τ for Policy H (red) and Policy S (blue), at median gate latency 30 ms (solid) and 150 ms (dashed).

Asset	baseline PnL (bps)	net impact (bps)		$\Delta(H-S)$ (bps)	agg. winner	S wins (of 10)
		H	S			
SPXUSD	−3454	+16419	+9250	+7168	H	1
USDJPY	+1538	−19580	−9683	−9897	S	10
XAUUSD	+25	+10402	+6409	+3993	H	3
ETHUSD	+9083	−116844	−54706	−62137	S	10

Table 7: Aggregate winner across the ten (m, τ) configurations, uninformed rejection. $\Delta > 0$ favours H. The magnitude shows how much the choice matters.

On positive-edge assets (USDJPY, ETHUSD) Policy S wins all ten configurations, since blocking on timeout discards positive expectancy while scaling down preserves it. On SPXUSD (negative edge, −3454 bps) H wins nine of ten, but this is the strategy’s own negative edge showing through a control that happens to remove exposure, not genuine risk management. XAUUSD (+25 bps, near zero) is ambiguous. H wins the aggregate but loses three configurations to S.

Table 8 decomposes the $m=30, \tau=10$ configuration into foregone profitable and avoided losing trades.

3.3 Tail behaviour in the stress windows

Trades are partitioned into three non-overlapping buckets, namely COVID (15 Feb–15 Apr 2020), instrument-own stress (top 5% realised volatility days, clustered), and rest. Table 9 decomposes Policy H’s net impact per regime and shows which policy wins within each bucket.

The edge-sign rule holds at the bucket level, not only the full sample. XAUUSD flips to S inside COVID (−1,487 bps) because the crash generates positive safe-haven edge, so blocking discards winners, yet recovers under own stress (+2,203 bps) where volatility spikes turn edge negative. ETHUSD flips to H in its own stress

Asset	n foregone	bps foregone	n avoided	bps avoided	net (bps)
SPXUSD	273	36 197	719	38 635	+2 437
USDJPY	269	19 849	646	17 818	−2 031
XAUUSD	238	34 980	683	35 948	+969
ETHUSD	239	166 754	668	150 922	−15 832

Table 8: Foregone winners vs. avoided losers under Policy H ($m=30$ ms, $\tau=10$ ms). The gate blocks more losers by count ($n_{\text{avoided}} > n_{\text{foregone}}$ everywhere, the strategy’s hit rate is below 50%), yet S still wins on positive-edge assets because the average winner is larger than the average loser. The dominance question is a bps argument, not a count argument.

Asset	COVID		own stress		rest	
	impact (bps)	W	impact (bps)	W	impact (bps)	W
SPXUSD	+714 (2.6%)	H	+665 (3.1%)	H	+1,059 (94.3%)	H
USDJPY	−1,129 (1.7%)	S	−524 (3.4%)	S	−378 (94.9%)	S
XAUUSD	−1,487 (1.4%)	S	+2,203 (3.3%)	H	+253 (95.3%)	H
ETHUSD	−3,910 (2.3%)	S	+3,060 (3.8%)	H	−14,982 (93.9%)	S

Table 9: Three-way stress partition ($m=30$ ms, $\tau=10$ ms, uninformed). Impact (bps) is Policy H’s net impact vs. the ungated baseline in that bucket. W shows which policy wins within the regime (higher net impact is better). Trade share (in parentheses) shows each bucket’s fraction of total orders.

bucket (+3,060 bps) for the same reason, but S dominates in the rest bucket (−14,982 bps) where the strategy retains positive edge. These reversals are exactly what the identity predicts. Across the ten (m, τ) configurations, only the two near-zero-edge assets (SPXUSD and XAUUSD) change policy winner. More precisely, Table 9 shows that the optimal policy is a function of the edge regime, not just the asset. ETHUSD flips sign across all three buckets, proving that instrument identity alone does not determine the choice. Replacing the uninformed reject with the informed variant does not change the aggregate winner on any instrument.

Failure modes. Policy H fails by discarding positive expectancy whenever timeout events are uninformative. Policy S fails by allowing reduced but unchecked exposure whenever the gate exceeds its latency budget. The choice is therefore between two measurable failure modes rather than between “safe” and “fast”, and both are quantified throughout this section.

3.4 Recommendation

There is no universally dominant timeout policy. The optimal choice is a function of the strategy’s edge sign, the timeout rate, and the stress regime. On this trade stream Policy S dominates on positive-edge instruments (USDJPY, ETHUSD) across all ten configurations, while Policy H dominates on negative-edge instruments (SPXUSD) across nine. XAUUSD (near-zero edge) is ambiguous.

The rule extends the Part B observation that latency is primarily a dispersion amplifier rather than a stable tax. Refusing to trade on a mechanical latency accident is only cheaper than trading small when the orders themselves carry no edge to preserve. The two policies cost almost the same on a healthy stream and differ mainly in how they treat an unhealthy one, and where Policy H wins the sharper response is to question whether the strategy should trade that instrument at all. The engineering decision is therefore not “which timeout policy is safer”, but whether timeout should remove exposure or preserve reduced exposure conditional on the measured edge of the underlying strategy.

4 Sensitivity of the conclusions

Three assumptions carry most of the weight above. First, resolution. The Part A interpolation and partial fill proxy, and Part C’s g_i repricing, all operate at M1 resolution, where a millisecond scale shift mostly lands in the same bar as the baseline, so the ETH exit slippage estimate and the H versus S margin at low timeout rates would both likely grow, not shrink, on tick data. Second, the latency distributions (Part B’s delay grid, Part C’s gate latency) are lognormal defaults chosen because the spec permits them without justification, not fitted to any observed venue behaviour, so decay and policy margins scale with a shape assumption rather than a measured one. Third, the sample dependency of specific findings. The reference strategy’s per instrument edge sign, on which the Policy S versus H recommendation is keyed, is an empirical property of this six year sample and not a structural one. Each of these can be tightened (tick data, a fitted latency model, a gate aware resimulation) without changing the qualitative conclusions.

References

Cartea, Á. and Sánchez Betancourt, L. (2022). Optimal execution with stochastic delay. *Finance and Stochastics* 27, 1–47.